

Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

- A. Decreased prolactin, decreased FSH, decreased LH
- B. Decreased prolactin, increased FSH, increased LH
- C. Increased prolactin, decreased FSH, increased LH
- D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>

** Finding reasoning paths : **

- 1 . Missed periods and spontaneous milky -white nipple discharge suggest a hormonal imbalance , possibly due to a pituitary tumor or hyperprolactinemia .
- 2 . Hyperprolactinemia is often associated with increased prolactin levels and may lead to galactorrhea and amenorrhea .
- 3 . Elevated prolactin levels can be caused by prolactin-secreting tumors or hyperpituitarism .
- 4 . Neurologic examination , including normal visual fields , does not suggest a hypothalamic or pituitary tumor .
- 5 . Increased prolactin could lead to decreased gonadotropin (FSH and LH) levels , causing anovulation and amenorrhea .

** Reasoning Process : **

- 1 . The patient has missed her last 3 periods , indicating amenorrhea , and has been experiencing spontaneous milky -white nipple discharge . These symptoms are indicative of hyperprolactinemia , which is often associated with increased prolactin levels .
- 2 . Hyperprolactinemia can be caused by several conditions , including prolactin-secreting tumors , medication side effects , or pituitary hyperplasia . Given the absence of neurologic abnormalities such as vision changes , a non-pituitary cause seems less likely .
- 3 . Elevated prolactin levels are known to suppress the pituitary gland 's ability to produce FSH and LH , which are gonadotropins responsible for ovulation and menstruation . This suppression leads to anovulation and amenorrhea .
- 4 . The milky -white discharge is a hallmark of galactorrhea , which is often linked to hyperprolactinemia .
- 5 . In hyperprolactinemia , the prolactin levels are elevated , while gonadotropin levels (FSH and LH) are decreased due to the negative feedback mechanism on the pituitary gland .

** Conclusion : **

The patient's laboratory results would most likely show increased prolactin levels and decreased FSH and LH levels . This pattern is consistent with hyperprolactinemia , which can be caused by factors such as a prolactin-secreting tumor or medications . The decreased gonadotropins explain the amenorrhea and potential galactorrhea . Therefore , the expected laboratory findings for this patient are increased prolactin , decreased FSH , and decreased LH .

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< answer >

Increased prolactin , decreased FSH , decreased LH

</answer>

